

Variation across speech and writing

Douglas Biber

Similarities and differences between speech and writing have been the subject of innumerable studies, but until now there has been no attempt to provide a unified linguistic analysis of the whole range of spoken and written registers in English. In this widely acclaimed empirical study, Douglas Biber uses computational techniques to analyse the linguistic characteristics of twenty-three spoken and written genres, enabling identification of the basic, underlying dimensions of variation in English.

In *Variation across speech and writing*, six dimensions of variation are identified through a factor analysis, on the basis of linguistic co-occurrence patterns. The resulting model of variation provides for the description of the distinctive linguistic characteristics of any spoken or written text and demonstrates the ways in which the polarization of speech and writing has been misleading, and thus enables reconciliation of the contradictory conclusions reached in previous research.

Variation across speech and writing

Variation across speech and writing

Douglas Biber

Department of English, Northern Arizona University



CAMBRIDGE
UNIVERSITY PRESS

CAMBRIDGE UNIVERSITY PRESS

Cambridge, New York, Melbourne, Madrid, Cape Town, Singapore,
São Paulo, Delhi, Dubai, Tokyo, Mexico City

Cambridge University Press

The Edinburgh Building, Cambridge CB2 8RU, UK

Published in the United States of America by

Cambridge University Press, New York

www.cambridge.org

Information on this title: www.cambridge.org/9780521425568

© Cambridge University Press 1988

This publication is in copyright. Subject to statutory exception
and to the provisions of relevant collective licensing agreements,
no reproduction of any part may take place without the written
permission of Cambridge University Press.

First published 1988

First paperback edition 1991

Reprinted 1995

A catalogue record for this publication is available from the British Library

Library of Congress Cataloguing in Publication Data

Biber, Douglas.

Variation across speech and writing / Douglas Biber.

p. cm.

Bibliography.

Includes index.

ISBN 0 521 32071 2

1. Language and languages – Variation. 2. Oral communication.

3. Written communication. I. Title.

PT120.V37B54 1988

001.54–dc19 87-38213

ISBN 978-0-521-32071-9 Hardback

ISBN 978-0-521-42556-8 Paperback

Cambridge University Press has no responsibility for the persistence or
accuracy of URLs for external or third-party internet websites referred to in
this publication, and does not guarantee that any content on such websites is,
or will remain, accurate or appropriate. Information regarding prices, travel
timetables, and other factual information given in this work is correct at
the time of first printing but Cambridge University Press does not guarantee
the accuracy of such information thereafter.

*For my parents,
Martha and Herb Biber*

Contents

<i>List of figures</i>	<i>page xi</i>
<i>List of tables</i>	<i>xiii</i>
<i>Acknowledgments</i>	<i>xv</i>
 Part I Background concepts and issues	 1
1 Introduction: textual dimensions and relations	3
2 Situations and functions	28
3 Previous linguistic research on speech and writing	47
 Part II Methodology	 59
4 Methodological overview of the study	61
5 Statistical analysis	79
 Part III Dimensions and relations in English	 99
6 Textual dimensions in speech and writing	101
7 Textual relations in speech and writing	121
8 Extending the description: variation within genres	170
9 Afterword: applying the model	199
 Appendix I Texts used in the study	 208
Appendix II Linguistic features: algorithms and functions	211
Appendix III Mean frequency counts of all linguistic features in each genre	246
Appendix IV Pearson correlation coefficients for all linguistic features	270
 <i>References</i>	 280
<i>Index</i>	<i>293</i>

Figures

	<i>page</i>
1.1 One-dimensional plot of four genres: nominalizations and passives	17
1.2 One-dimensional plot of four genres: first and second person pronouns and contractions	17
1.3 Two-dimensional plot of four genres	18
1.4 One-dimensional plot of four genres: third person pronouns and past tense verbs	18
2.1 Oral and literate situational characteristics of four genres	46
5.1 Scree plot of eigenvalues	83
5.2 Mean scores of Dimension 2 for each of the genres	96
7.1 Mean scores of Dimension 1 for each of the genres	128
7.2 Mean scores of Dimension 2 for each of the genres	136
7.3 Mean scores of Dimension 3 for each of the genres	143
7.4 Mean scores of Dimension 4 for each of the genres	149
7.5 Mean scores of Dimension 5 for each of the genres	152
7.6 Mean scores of Dimension 6 for each of the genres	155
7.7 Plot of the textual relations among seven genres, highlighting face-to-face conversation, personal letters, spontaneous speeches, and broadcasts	165
7.8 Plot of the textual relations among seven genres, highlighting official documents, professional letters, and general fiction	166
8.1 Spread of scores along Dimension 1 for selected genres	172
8.2 Spread of scores along Dimension 2 for selected genres	173
8.3 Spread of scores along Dimension 3 for selected genres	174
8.4 Spread of scores along Dimension 4 for selected genres	175
8.5 Spread of scores along Dimension 5 for selected genres	176
8.6 Spread of scores along Dimension 6 for selected genres	177
8.7 Mean scores of Dimension 1 for selected sub-genres	185
8.8 Mean scores of Dimension 2 for selected sub-genres	186
8.9 Mean scores of Dimension 3 for selected sub-genres	187
8.10 Mean scores of Dimension 4 for selected sub-genres	188
8.11 Mean scores of Dimension 5 for selected sub-genres	189
8.12 Mean scores of Dimension 6 for selected sub-genres	190

Tables

1.1	Frequency counts for texts 1.1, 1.2, and 1.3	<i>page</i> 15
1.2	Additional frequency counts for texts 1.1, 1.2, 1.3, and 1.4	19
2.1	Components of the speech situation	30-1
2.2	Functions of linguistic features	35
2.3	Major situational parameters distinguishing between typical speaking and writing	39
4.1	Steps in the analysis	
4.2	Distribution of texts across 23 genres	64
4.3	List of sub-genres used in the study	67
4.4	Features used in the analysis	69
4.5	Descriptive statistics for the corpus as a whole	73-5
5.1	First eleven eigenvalues of the unrotated factor analysis	83
5.2	Inter-factor correlations	84
5.3	Rotated factor pattern for the 7 factor solution	86-7
5.4	Summary of the factorial structure	89-90
5.5	Factor 3 of the 6 factor solution	91
6.1	Summary of the factorial structure	102-3
6.2	Summary of the factorial structure of 41 linguistic features	117
7.1	Descriptive dimension statistics for all genres	122-5
7.2	F and correlation scores for the six textual dimensions	127
8.1	Descriptive statistics for specialized sub-genres	181-4

Acknowledgments

I first became interested in the relationship between speech and writing during my dissertation research, and several of the methodological tools used in the present study were developed at that time. More recently, my dissertation advisors have become my academic colleagues, so that I owe them thanks for both their earlier guidance and their continuing criticism and support. In particular, Edward Finegan, Elinor Ochs, Edward Purcell, and June Shoup have helped me from the earliest stages of this research. Ed Finegan deserves special thanks, since he has been intimately involved in all stages of the research study presented here. As my dissertation chair, he helped conceptualize the methodological approach used here; for example, Ed was the first to suggest that I use computational tools to analyze spoken and written texts, and he went to considerable effort to help obtain computerized text corpora for analysis. He has worked with me on numerous papers and research studies, and his criticisms have ranged from writing style and rhetorical organization to theoretical interpretation and conceptual presentation of results. Finally, over the last several years Ed and I have collaborated on many related studies of variation in English. It is not possible to isolate individual effects of these contributions; they have all influenced the final form of the present study.

Several other colleagues helped directly with the present book. Pat Clancy and Gunnel Tottie both read the manuscript and made especially detailed and helpful comments. Bill Grabe and Niko Besnier helped me through numerous conversations as well as their comments on the manuscript. Larry Ploetz answered numerous computer-related questions during the development of the linguistic analysis programs used here.

Other colleagues at U.S.C. did not help directly with this book but offered their friendship and support; of these, I want to single out Steve Krashen, Larry Hyman, and Joseph Aoun.

Finally, this book would not have been possible without the continued support and understanding of my wife, Teresa, and my children, David and Martha. Although their contributions are less tangible, they are in many ways greater than any of the others.

Part I: Background concepts and issues

1 Introduction: textual dimensions and relations

1.1 Introduction

A considerable body of research in the humanities and social sciences has dealt with the similarities and differences between speech and writing. Work in history, sociology, anthropology, psychology, education, comparative literature, and linguistics has described ways in which the choice between speech and writing is closely related to developments in other social institutions. For example, the development of widespread alphabetic literacy in ancient Greece was probably a catalyst for other social and intellectual developments there. Widespread literacy enabled a fuller understanding and participation by citizens in the workings of government, which might have promoted a democratic form of government in which citizens play a relatively active role. Literacy enabled a permanent, accurate record of ideas and the possibility of knowledge without a living 'knower'. As such it probably aided in the transition from 'myth' to 'history' and the development of critical attitudes towards knowledge. Prior to literacy and a permanent record of beliefs and knowledge, a society can alter its beliefs and not be faced with the possibility of a contradiction; competing ideas which evolve slowly over generations will be accepted as equally factual when there is no contradictory record of earlier ideas. Written records, however, force us to acknowledge the contradictory ideas of earlier societies and thus to regard knowledge with a critical and somewhat skeptical attitude. For example, we know that earlier societies believed that the earth was flat, because these beliefs are permanently recorded in writing. The permanency of writing thus confronts us with the incorrect 'knowledge' of earlier generations and thereby fosters a generally critical attitude towards knowledge.

The permanency of writing also enables the dissection of texts, so that ideas can be critically examined in the abstract and the logical relations among ideas can be discussed. Literacy enables language itself to be the

4 *Background concepts and issues*

object of inquiry. These possibilities helped foster the development of philosophy as we know it. Similarly, the use of literacy aided the development of new literary forms. Oral literature tends to be poetic, because poetic forms are more easily memorized and transmitted from one generation to the next. The permanency of written texts enables an accurate transmission of any literary form, enabling experimentation with non-poetic types. Although the transition to literacy did not by itself cause any of these intellectual or social developments, it seems to have been an important catalyst.¹

The transition to literacy seems to have important consequences for individuals as well as societies. Some researchers have claimed that radically different thought processes are enabled by literacy. In particular, it has been claimed that abstract, 'decontextualized' thought depends on literacy, so that non-literate individuals can think in only concrete, contextualized ways. This claim is difficult to evaluate because literacy in Western culture is always confounded with formal education, and thus intellectual differences between literate and non-literate individuals might be due to either the acquisition of literacy or the educational process itself. Research in West Africa by Scribner and Cole (1981) has helped to isolate the effects of literacy from those of formal schooling. The Vai people, who live in Liberia, have developed an indigenous writing system that is used only for traditional, non-academic purposes. Vai literates are taught how to write on an individual basis apart from any other formal schooling. Other members of this tribe become literate in Arabic to study the Quran, or in English by attendance at government schools. Scribner and Cole found that there are specific intellectual abilities which are enhanced by each type of literacy, depending on the particular functions served. For example, Quranic literacy among the Vai greatly enhances memorization abilities because beginning students learn to 'read' the Quran without understanding, and they use their readings to help memorize large portions of the text. Consequences of this type are minor and quite specific to different types of literacy; Scribner and Cole found no global intellectual consequences of literacy apart from the influence of formal schooling.

Although the primary intellectual consequences of literacy are subsumed under those of formal education, there is obviously a very close relationship between school success and literacy. Children who fail at

¹ See Goody and Watt (1963), Goody (1977), Stubbs (1980), Ong (1982), and Street (1984) for further discussion of the social and cultural consequences of literacy.

reading and writing fail at school; children who fail at school don't learn how to read. It is difficult to establish a causal relationship here, but schooling is inextricably bound to literacy in Western culture. Several researchers have investigated the acquisition of literacy, and its relation to school success, in Western societies. Some studies describe the problems caused by reliance on spoken language strategies in the compositions of basic writers. Other researchers, such as Heath and Wells, emphasize that many successful students acquire the language-use strategies associated with literacy long before they can actually read and write, and that these strategies are crucially important to literacy acquisition and the types of language use required for school tasks. These patterns of language use can be conveyed by reading to children in the home, but they are further developed by decontextualized spoken interactions; for example, hypothetical discussion of what a storybook character might have done in a particular situation. Students who begin school 'literate', in the sense that they already realize that language can be used for abstract, decontextualized purposes, are the ones who adapt most easily to the requirements of Western education.

Studies similar to these, which look for social or intellectual correlates of writing as distinct from speaking, are found throughout the humanities and social sciences. Given this wide range of interest, it might be expected that the linguistic characteristics of spoken and written language have been thoroughly analyzed. There have, in fact, been many linguistic studies of speech and writing, but there is little agreement on the salient characteristics of the two modes. The general view is that written language is structurally elaborated, complex, formal, and abstract, while spoken language is concrete, context-dependent, and structurally simple. Some studies, though, have found almost no linguistic differences between speech and writing, while others actually claim that speech is more elaborated and complex than writing.

There has also been considerable disagreement concerning the need for a linguistic comparison of speech and writing. Historically, academics have regarded writing, in particular literary works, as the true form of language, while speech has been considered to be unstable, degenerate and not worthy of study. In the nineteenth century this situation began to change when linguists such as Grimm in Germany began to study speech in its own right. The development of phonetics as a separate discipline in Britain, primarily through the work of Henry Sweet and Daniel Jones, further encouraged linguists to study speech. These research trends,

6 *Background concepts and issues*

however, did not result in linguistic comparisons of speech and writing. Rather, by the early twentieth century, linguists uniformly regarded speech as primary and writing as a secondary form of language derived from speech; thus only speech was considered worth serious linguistic analysis. This bias can be traced from the time of Sapir up to the present, for example:

Sapir: *writing is 'visual speech symbolism' (1921:19–20)*

Bloomfield: *'writing is not language, but merely a way of recording language by visible marks' (1933:21)*

Hall: *'speech is fundamental and writing . . . only a secondary derivative' (1964:8–9)*

Postal: *'writing is a crude way of representing linguistic structure rather than a sign system with a direct relation to the world' (1966:91, n. 20)*

Fillmore: *written communication is 'derivative of the face-to-face conversational norm' (1981:153)*

Aronoff: *notes 'the undoubtedly correct observation that spoken language is "true" language, while written language is an artifact' (1985:28)*

Assuming this secondary, derivative nature of written language, there was no motivation within structural linguistics for comparison of speech and writing.

Although the bias that speech is primary over writing has been extremely important in guiding research efforts within linguistics, it has not been widely accepted outside of linguistics. In fact, the historical view that written, literary language is true language continues as the dominant lay perception to the present time. Our children need to study English at school, which includes written composition and the prescriptive rules of writing, not speech. We criticize immigrant children for not knowing 'English' when they are relatively fluent in a conversation; the problem is that they are not literate in English. We expect our grammars and dictionaries to present the correct forms of written language; when dictionaries present both literate and colloquial vocabulary, they are severely criticized for destroying the standards of English, as happened to

Webster's Third, which has been described as a 'disappointment', 'a scandal and a disaster' (see discussion in Finegan 1980). In our business, legal, and political systems, written commitments are binding and 'real' while spoken commitments are often ignored. As teachers, we explain to children that words like *know* have a silent [k], and words like *doubt* have a silent [b]. Sometimes we even change our pronunciation to reflect an unusual spelling; for example, *often* is now frequently pronounced with a [t], and *palm* with an [l], although these segments were not pronounced at some earlier stages of English. Thus, although speech is claimed to have linguistic primacy, writing is given social priority by most adults in Western cultures.²

Even within structural linguistics, researchers have not been entirely consistent regarding the primacy of speech. In particular, there has been a gap between theory and practice in recent syntactic research. In theory, writing is disregarded as secondary and derivative from speech. In practice, however, speech is also disregarded as unsystematic and not representative of the true linguistic structure of a language. This view is especially prominent within the generative-transformationalist paradigm, where grammatical intuitions are the primary data to be analyzed. Although these intuitions are typically collected by means of verbal elicitation, they are in many respects more like writing than speech. Thus the data for analysis within this paradigm deliberately exclude performance errors of 'actual speech', dialect, and register variation, and any linguistic features that depend on a discourse or situational context for interpretation. Although these data are not taken from actual speech or actual writing, they are much closer to stereotypical writing than speech in their form.

All of these perspectives regard either speech or writing as primary and representative of 'true' language; none grants independent status to both speech and writing. However, given the range of arguments on both sides of this issue, it might well be the case that neither speech nor writing is primary; that they are rather different systems, both deserving careful analysis. This is in fact the view advocated by Hymes and other researchers studying communicative competence. That is, in addition to the knowledge that all speakers have about the grammatical structure of

² The discussion here owes much to Stubbs (1980). Other works dealing with the primacy of speech or writing include Householder (1971), Vachek (1973), Basso (1974), Schafer (1981), Akinnaso (1982), and Stubbs (1982).

8 *Background concepts and issues*

their language, speakers also have extensive knowledge about the use of their language. The former knowledge is grammatical competence, which includes the traditional areas of phonology, syntax, and semantics. The latter knowledge is known as 'communicative competence', and includes formal knowledge of the range of speech-act variation, dialect variation, and register variation, as well as knowledge of when these different linguistic forms are appropriate. Grammatical competence is concerned with the linguistic structure of 'grammatical' utterances; communicative competence is concerned with the form and use of all language – both speech and writing. Within this framework, neither speech nor writing needs to be considered primary to the exclusion of the other. Rather, both require analysis, and the linguistic comparison of the two modes becomes an important question.

Of course, in terms of human development, speech has primary status. Culturally, humans spoke long before they wrote, and individually, children learn to speak before they read or write. All children learn to speak (barring physical disabilities); many children do not learn to read and write. All cultures make use of spoken communication; many languages do not have a written form. From a historical and developmental perspective, speech is clearly primary.

Once a culture has developed written communication, however, there is no reason to regard writing as secondary within that context. It has long been known that cultures exploit variation in linguistic form for functional purposes. For example, variation between lexical items such as *lorry* and *truck* functions to mark geographical differences; variation between pronunciations such as [ka:] versus [kar] and [dis] versus [dis] functions to mark social differences; variation in address terms, such as *Dr. Jones* versus *Sue*, functions to mark the formality of the situation and the social role relationship between speaker and listener. Similarly, once a culture develops a written form in addition to a spoken form, the two modes come to be exploited for different communicative purposes. Although either speech or writing *can* be used for almost any communicative need, we do not in fact use the two forms interchangeably. Rather, depending on the situational demands of the communicative task, we readily choose one mode over the other. Usually this choice is unconscious, since only one of the modes is suitable or practical. For example, we have no trouble choosing between leaving a note for someone or speaking to the person face-to-face; the situation dictates the

mode of communication. Similarly, we have no problem deciding between writing an academic exposition for an audience and addressing the audience by means of a spoken lecture. We could in fact write a lecture or a note to a physically present audience, but this would take more effort and time than required, and it would fail to take advantage of the opportunities for interaction. Conversely, speaking a lecture or a note to an addressee who is separated by time or place is usually not possible at all; apart from the use of telephones and tape recorders, the written mode is required in situations of this type. These simple examples illustrate the fact that the two modes of communication have quite different strengths and weaknesses, and they therefore tend to be used in complementary situations. From this perspective, neither can be said to be primary; they are simply different. The linguistic characteristics of each mode deserve careful attention, and the relationship between the two modes must be investigated empirically rather than assumed on an *a priori* basis.

1.2 Dimensions and relations

In the present book, spoken and written texts are compared along 'dimensions' of linguistic variation. Researchers have considered texts to be related along particular situational or functional parameters, such as formal/informal, interactive/non-interactive, literary/colloquial, restricted/elaborated. These parameters can be considered as dimensions because they define continuums of variation rather than discrete poles. For example, although it is possible to describe a text as simply formal or informal, it is more accurate to describe it as more or less formal; formal/informal can be considered a continuous dimension of variation.

I will illustrate the concept of 'dimension' in this section by analysis of a few linguistic features in four texts. This illustration greatly oversimplifies the linguistic character of the dimensions actually found in English. Chapters 5–8 present a full analysis based on the distribution of 67 linguistic features in 481 texts. The discussion here thus provides a conceptual description of dimensions, rather than actually describing the complex patterns of variation in English speech and writing.

Following are two quite distinct text samples, which differ along several dimensions. Readers should identify some of the differences between them before proceeding to the following discussion.

10 Background concepts and issues

Text 1.1: Conversation – comparing home-made beer to other brands

A:	<i>I had a bottle of ordinary Courage's light ale, which I</i>	1
	<i>always used to like, and still don't dislike, at Simon</i>	2
	<i>Hale's the other day –</i>	3
	<i>simply because I'm, mm, going through a lean period at</i>	4
	<i>the moment waiting for this next five gallons to be ready,</i>	5
	<i>you know.</i>	6
B:	<i>mm</i>	7
A:	<i>It's just in the bottle stage. You saw it the other night.</i>	8
B:	<i>yeah</i>	9
A:	<i>and, mm I mean, when you get used to that beer, which</i>	10
	<i>at its best is simply, you know, superb, it really is.</i>	11
B:	<i>mm</i>	12
A:	<i>you know, I've really got it now, really, you know, got</i>	13
	<i>it to a T.</i>	14
B:	<i>yeah</i>	15
A:	<i>and mm, oh, there's no, there's no comparison. It tasted</i>	16
	<i>so watery, you know, lifeless.</i>	17
B:	<i>mm</i>	18

Text 1.2: Scientific exposition

<i>Evidence has been presented for a supposed randomness in</i>	1
<i>the movement of plankton animals. If valid, this implies that</i>	2
<i>migrations involve kineses rather than taxes (Chapter 10).</i>	3
<i>However, the data cited in support of this idea comprise</i>	4
<i>without exception observations made in the laboratory.</i>	5

Text 1.1 is taken from an ordinary, face-to-face conversation between friends. It represents the type of communication that we all experience every day. Text 1.2 is much more specialized, coming from a scientific exposition. In contrast to the conversation, relatively few speakers of English commonly read texts like 1.2, and an extremely small proportion are expected to write texts of this type. We might thus distinguish texts 1.1 and 1.2 on a dimension of common versus specialized.

These texts might also be contrasted on a dimension of unplanned versus planned. In text 1.1, speaker A talks without careful planning. At one point he switches topic in the middle of a sentence – in line 10, he begins a thought with *when you get used to that beer*, and two utterances later, in line 16, he completes the sentence with *there's no comparison*; in between these two utterances he notes that his homemade beer is superb when made properly (lines 10–11), and that he really knows how to make the brew now (lines 13–14). Text 1.2 is quite different, having a very careful logical progression indicating careful planning. An idea is presented in lines 1–2, implications of the idea are given in line 3, and the idea is qualified in lines 4–5. This logical progression continues in the rest of text 1.2.

There are several other dimensions that these two texts could be compared along. For example, text 1.1 is interactive while text 1.2 is not; in text 1.1, speaker A refers directly to himself and to speaker B (*I* and *you*), and speaker B responds to A. Text 1.1 is dependent on the immediate situation to a greater extent than text 1.2; in text 1.1, speaker A assumes that B can identify *Simon Hale's* (line 3), *the other day* (line 3), *this next five gallons* (line 5), and *the other night* (line 8). The speaker in text 1.1 displays his feelings enthusiastically and emphatically, while the feelings of the writer in text 1.2 are less apparent; speaker A in text 1.1 repeatedly emphasizes his point with *really*, *simply*, and *you know* (lines 4, 6, 11, 13, 17).

When only two texts are compared, these parameters seem to be dichotomies. If we add a third text, however, we begin to see that these parameters define continuous dimensions. Thus, consider text 1.3 below:

Text 1.3: Panel discussion – discussing corporal punishment as a deterrent to crime

W: *But Mr. Nabarro, we know that you believe this.*

L: *quite*

W: *The strange fact is, that you still haven't given us a reason for it. The only reason you've given for us is, if I may spell it out to you once more, is the following:
the only crime for which this punishment was a punishment,
after its abolition, decreased for eleven years.*

You base on this the inference that if it had been applied to crimes it never had been applied to, they wouldn't have increased.

Now this seems to me totally tortuous.

Text 1.3 is intermediate between texts 1.1 and 1.2 with respect to the dimensions outlined above. Text 1.3 is certainly not a common everyday communication like text 1.1, but it is not as specialized as text 1.2; text 1.3 is relatively unplanned, but it is more carefully organized than text 1.1; text 1.3 is interactive, but not to the extent of text 1.1; text 1.3 shows little dependence on the immediate situation, but more so than text 1.2; and for the most part, the main speaker in text 1.3 does not reveal his own feelings, although they are more apparent than those of the writer in text 1.2. Text 1.3 is more like text 1.1 with respect to some of these dimensions, and more like text 1.2 with respect to others. However, it has an intermediate characterization with respect to texts 1.1 and 1.2 on each dimension, indicating that these are continuous parameters rather than simple dichotomies.

To this point, we have discussed the notion of dimension from a situational or functional point of view. It is also possible to discuss this notion from a strictly linguistic perspective. In the same way that texts can be described and compared in terms of their situational characterization, there are dimensions that compare texts in terms of their linguistic characterization, e.g., nominal versus verbal, or structurally complex versus structurally simple. Thus consider texts 1.1, 1.2, and 1.3 again. A general impression of text 1.1 is that it is verbal rather than nominal (i.e., many verbs, few nouns) and that it is structurally simple (e.g., little phrasal or clausal elaboration). Text 1.2, on the other hand, seems to be extremely nominal and structurally complex, while text 1.3 seems to have a linguistic characterization between these two. Several questions arise, though: (1) What evidence can we give to support these linguistic impressions? (2) Do these characterizations represent a single linguistic dimension, or two dimensions, or more than two? How can a researcher determine how many linguistic dimensions are required to account for the variation among a set of texts? (3) Are there other linguistic dimensions that are not represented by the above linguistic impressions? If so, how can they be discovered?

I develop an overall empirical approach in the present book that addresses these questions (cf. Section 1.3, Section 3.5, and Chapter 4).

The raw data of this approach are frequency counts of particular linguistic features. Frequency counts give an exact, quantitative characterization of a text, so that different texts can be compared in very precise terms. By themselves, however, frequency counts cannot identify linguistic dimensions. Rather, a linguistic dimension is determined on the basis of a consistent co-occurrence pattern among features. That is, when a group of features consistently co-occur in texts, those features define a linguistic dimension. It should be noted that the direction of analysis here is opposite from that typically used in studies of language use. Most analyses begin with a situational or functional distinction and identify linguistic features associated with that distinction as a second step. For example, researchers have given priority to functional dimensions such as formal/informal, restricted/elaborated, or involved/detached, and subsequently they have identified the linguistic features associated with each dimension. In this approach, the groupings of features are identified in terms of shared function, but they do not necessarily represent linguistic dimensions in the above sense; that is these groupings of features do not necessarily represent those features that co-occur frequently in texts. The opposite approach is used here: quantitative techniques are used to identify the groups of features that actually co-occur in texts, and afterwards these groupings are interpreted in functional terms. The linguistic dimension rather than functional dimension is given priority.

This approach is based on the assumption that strong co-occurrence patterns of linguistic features mark underlying functional dimensions. Features do not randomly co-occur in texts. If certain features consistently co-occur, then it is reasonable to look for an underlying functional influence that encourages their use. In this way, the functions are not posited on an a priori basis; rather they are required to account for the observed co-occurrence patterns among linguistic features.

In fact, there are several unaddressed issues surrounding dimensions identified on functional bases. Although many functional dimensions have been proposed in recent years, few researchers have attempted to relate them to one another or rank them in importance. Consider the following partial list of functional dimensions: informal/formal, restricted/elaborated, contextualized/decontextualized, involved/detached, integrated/fragmented, abstract/concrete, colloquial/literary. Are these all separate dimensions? Do some of them overlap? Are they all equally important? Are they all well-defined in terms of their linguistic

characterization? The approach used here begins to answer these questions. By defining 'dimension' from a strictly linguistic perspective, it is possible to identify the set of dimensions required to account for the linguistic variation within a set of texts. Each dimension comprises an independent group of co-occurring linguistic features, and each co-occurrence pattern can be interpreted in functional terms. The result is an empirical assessment of how many independent dimensions there are; an assessment of which functions are independent and which are associated with the same dimension; and an assessment of the relative importance of different dimensions.

The discussion can be made more concrete by considering some frequency counts in texts 1.1, 1.2, and 1.3. In Table 1.1, I list the frequencies for four linguistic features: passive constructions (including post-nominal modifiers, e.g., *the data [which are] cited*), nominalizations, first and second person pronouns, and contractions. The table includes the raw frequency count and the frequency per 100 words; I use the frequency counts normalized to a text of 100 words to compare the three texts.³

The conversational text (1.1) and scientific text (1.2) are quite different with respect to these linguistic features. The scientific text has almost seven passives per 100 words and eleven nominalizations per 100 words; the conversation has no passives and less than one nominalization per 100 words. Assuming that these two texts are representative of their kind, their frequency counts indicate that passives and nominalizations tend to co-occur and thus belong to the same linguistic dimension – when a text has many passives, it also has many nominalizations, as in the scientific text; when a text has few passives, it also has few nominalizations, as in the conversational text. Similarly, these two texts indicate that first and second person pronouns and contractions belong to the same dimension – when a text has many first and second person pronouns, it also has many contractions, as in the conversational text; when a text has few first and

³ Raw frequency counts cannot be used for comparison across texts because they are not all the same length. That is, long texts will tend to have higher frequencies simply because there is more opportunity for a feature to occur; in these cases, the higher count does not indicate a more frequent use of the feature. Comparing the frequency per 100 words eliminates this bias. These normalized frequencies are computed as follows:

$$(\text{actual frequency count} \div \text{total words in text}) \times 100$$

For example, the normalized frequency of contractions in text 1.1 is:

$$(6 \div 118) \times 100 = 5.1$$